

ALAZ

Title: Regional Machine Learning and Climate Index-Based Forest Fire Early Warning Decision Support System: A Turkey Application

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1. ABSTRACT

Turkey has experienced severe escalations in wildfire frequency. Historically, fire prediction relied on the Canadian Fire Weather Index (FWI) System, which quantifies combustion-favorable weather but inherently fails to model actual ignition probabilities, anthropogenic influences, and vegetation phenology. This study presents ALAZ, a regional forest fire early warning system engineered for Turkey. The forested terrain is tessellated into 235,621 H3 hexagonal grids (~0.74 km²) and partitioned into Mediterranean (MED), Continental (CON), and Black Sea (BLK) macro-regions based on EEA 2016 zoning. Addressing extreme class imbalance (~1:100,000 daily prevalence) through six physically-meaningful negative sampling strategies, the system trains region-specific XGBoost classifiers. These models integrate 36 spatial-temporal features, including ERA5 meteorology, FWI sub-indices, MODIS NDVI, SRTM topography, and OpenStreetMap anthropogenic factors. Calibrated via Platt Scaling to ensure statistical reliability (Brier Scores < 0.088), the models are deployed through a high-performance web-based Decision Support System. This architecture performs real-time NetCDF processing and XGBoost inference, generating dynamic, EFFIS-compliant risk visualizations for operational management.

2. REVIEW OF LITERATURE

Forest fire prediction has evolved from traditional indices (FWI, KBDI) [4, 8]—which quantify fuel drying but ignore ignition probability—to machine learning (ML) frameworks. Recent studies validate ML's superiority: Yang et al. [1] proved XGBoost's efficacy in capturing complex interactions, while Sinato & Rivas [2] pioneered H3 hexagonal grids to minimize spatial distortions. Focusing on Turkey, Iban & Aksu [3] utilized SHAP for active fire interpretation, and Atalay et al. [8] mapped danger via EFFIS standards. Temporal and GIS-integrated works are evident in models by Lee et al. [5] and Dong et al. [4], alongside mapping frameworks in Algeria [6], Australia [9], and Morocco [7]. Research Gap: Despite advancements, literature mostly trains monolithic models, ignoring that fire drivers differ radically across biogeographical zones (e.g., Black Sea moisture vs. Mediterranean drought defaults). Moreover, studies output uncalibrated ML margin scores as "probabilities," limiting operational utility. ALAZ bridges these gaps via strict biogeographical regionalization and empirically calibrated outputs.

3. METHODOLOGY

3.1. Spatial Tessellation and Regionalization

The study area utilizes CORINE Land Cover 2018 data, filtering five forest/shrub classes (311, 312, 313, 323, 324) comprising 23.8 million pixels. To eliminate boundary distortions inherent in raster data, the area was tessellated using Uber's H3 Hierarchical Hexagonal Engine (Resolution 8), resulting in 235,621 unique forest grid cells. These cells were strictly partitioned into three macro-regions using the European Environment Agency (EEA) Biogeographical Regions 2016 dataset: Mediterranean (MED, 52%), Black Sea (BLK, 27%), and Continental (CON, 21%).

3.2. Data Integration and Feature Engineering

A historical database of 15,435 unique fire events (2008–2025) was curated by spatially deduplicating NASA FIRMS (MODIS/VIIRS) and EFFIS polygons. The models ingest a 36-dimensional physical feature space spanning five domains: Meteorology (ERA5-Land) supplies temperature, humidity, wind, VPD, and lag precipitation; the Fire Weather Index (Van Wagner) provides grid-level FFM, DMC, DC, ISI, BUI, and FWI; Phenology (MODIS Terra MOD13Q1) tracks monthly NDVI vegetation stress; Topography (SRTM 30m DEM) defines elevation, slope, northness, and eastness; and Anthropogenic Variables (OpenStreetMap) calculate road density, settlement proximity, and the critical temporal metric of days_since_last_fire.

$$m = 147.2 \cdot \frac{101 - FFM}{59.5 + FFM}$$
$$f(W) = e^{0.05039 - W}$$
$$f(F) = 91.9 \cdot e^{-0.1386 \cdot m} \cdot \left[1 + \frac{m^{5.31}}{4.93 \times 10^7} \right]$$
$$ISI = 0.208 \cdot f(W) \cdot f(F)$$
$$BUI = \begin{cases} \frac{0.8 \cdot DMC \cdot DC}{DMC + 0.4 \cdot DC}, & DMC \leq 0.4 \\ DMC - \left[1 - \frac{0.8 \cdot DC}{DMC + 0.4 \cdot DC} \right] \cdot [0.92 + (0.0114 \cdot DMC)^{1.7}], & DMC > 0.4 \end{cases}$$
$$f(D) = \begin{cases} 0.626 \cdot BUI^{0.809} + 2, & BUI \leq 80 \\ \frac{1000}{25 + 108.64 \cdot e^{-0.023 \cdot BUI}}, & BUI > 80 \end{cases}$$
$$B = 0.1 \cdot ISI \cdot f(D)$$
$$FWI = \begin{cases} B, & B \leq 1 \\ \exp[2.72 \cdot (0.434 \cdot \ln(B))^{0.647}], & B > 1 \end{cases}$$

FWI Formulas

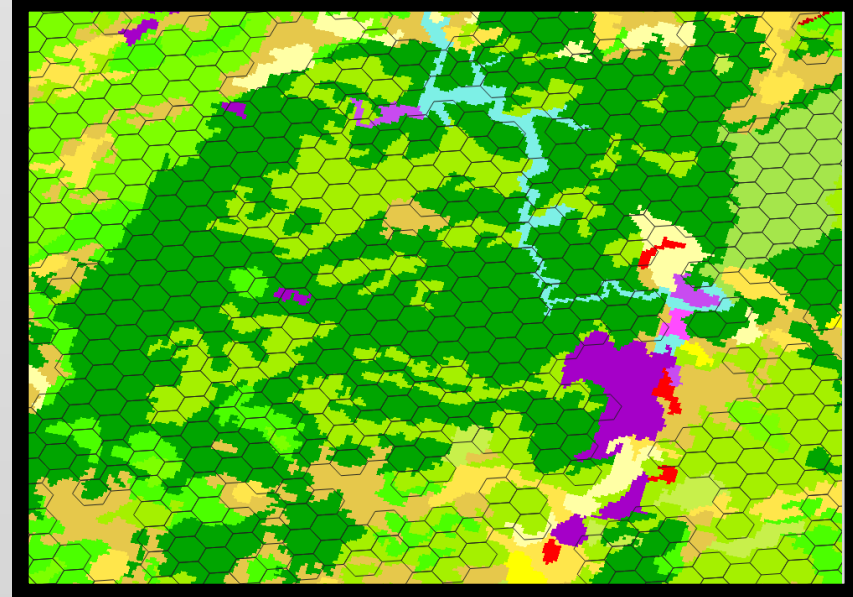


3.3. Negative Sampling Architecture

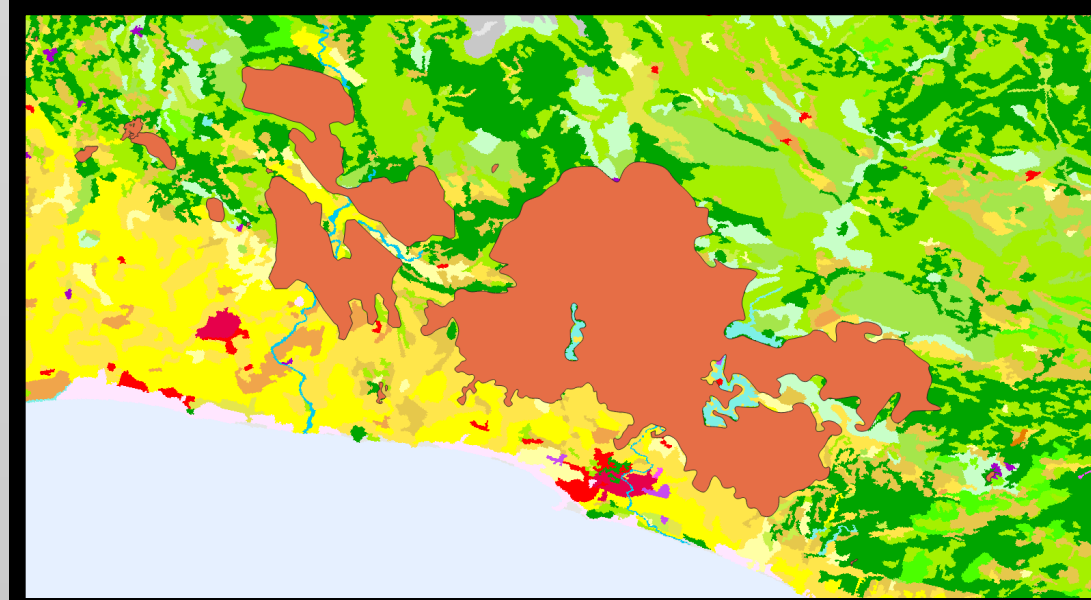
True daily fire prevalence is approximately 1:100,000. Randomly sampling negative (non-fire) data leads to the "Accuracy Paradox," where models superficially learn to predict "no fire" without capturing pre-ignition behaviors. To enforce rigorous learning, we engineered six physically meaningful negative profiles: Opposite Season (±180 days), Breaking Point (~2/~5 days prior to fire), Spatial Neighbor (adjacent cells), Intra-Season (±30 days), Historical Recurrence (~365 days), and Regional Reference (distant cells). Across 120 experimental trials, optimized negative strategies were identified for each region.

3.4. Model Training & Calibration

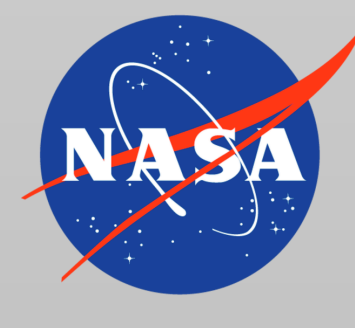
To strictly prevent temporal data leakage, an absolute chronological split was enforced: Train (2008–2021), Validate (2022), and Test (2023–2025). Regional XGBoost Gradient Boosting classifiers were hyperparameter-tuned using the Optuna Bayesian optimization framework. Crucially, to transform raw margin outputs into true operational probabilities, models were subjected to Platt Scaling (Sigmoid Calibration) on the validation set, ensuring exact risk threshold reliability.



H3 Grids



2021 Antalya Manavgat Burned Polygons



4. RESULTS AND FINDINGS

4.1. Regional Model Performance

on Unseen Test Data (2023–2025)
The models were evaluated strictly on three years of completely unseen data, proving high robustness against temporal bias. The implementation of Platt Scaling achieved Brier Scores under 0.088, signifying that the generated risk percentages statistically match real-world occurrence frequencies.

--- MED(: 0.22) ---
Accuracy : % 77.63
Precision : % 25.58
Recall : % 57.30
(Bal.Acc) : % 68.68
Specificity : % 80.06
AUC : 0.759
Brier: 0.088
F1: 0.354

4.2. The Impact of the Anthropogenic Factor

A comparative analysis isolating the impact of human features (road_density, dist_settlement, days_since_last_fire) proved that wildfire risk is inherently tied to human accessibility. The addition of anthropogenic metrics uniformly enhanced AUC metrics across the board (MED: +0.013, CON: +0.045, BLK: +0.048), decisively proving that weather alone is insufficient for predicting modern fire outbreaks.

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Orman Yangını Erken Uyarı Sistemi

Türkiye Geneli 235K H3 Grid Dinamik Ahtıyapı Sistemine hoşgeldiniz. Lütfen analiz edilecek zamanı ve modeli belirleyin.

Başlangıç Tarihi

20.07.2024

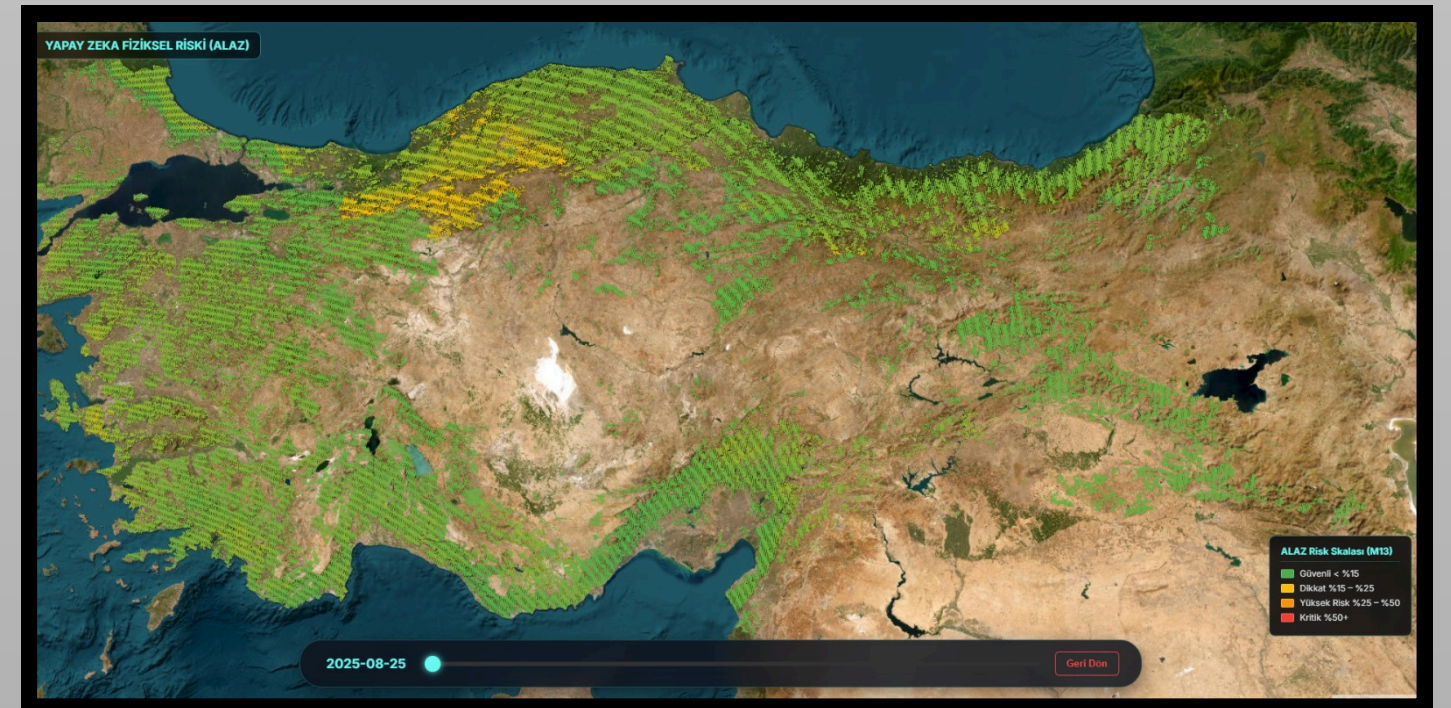
Bitiş Tarihi (Maks +7 Gün)

23.07.2024

Hesaplama Modu:

3. Karşılaştırma Modu (FWI vs ALAZ Kıyaslaması - Tek Günlük)

SİMÜLASYONU BAŞLAT

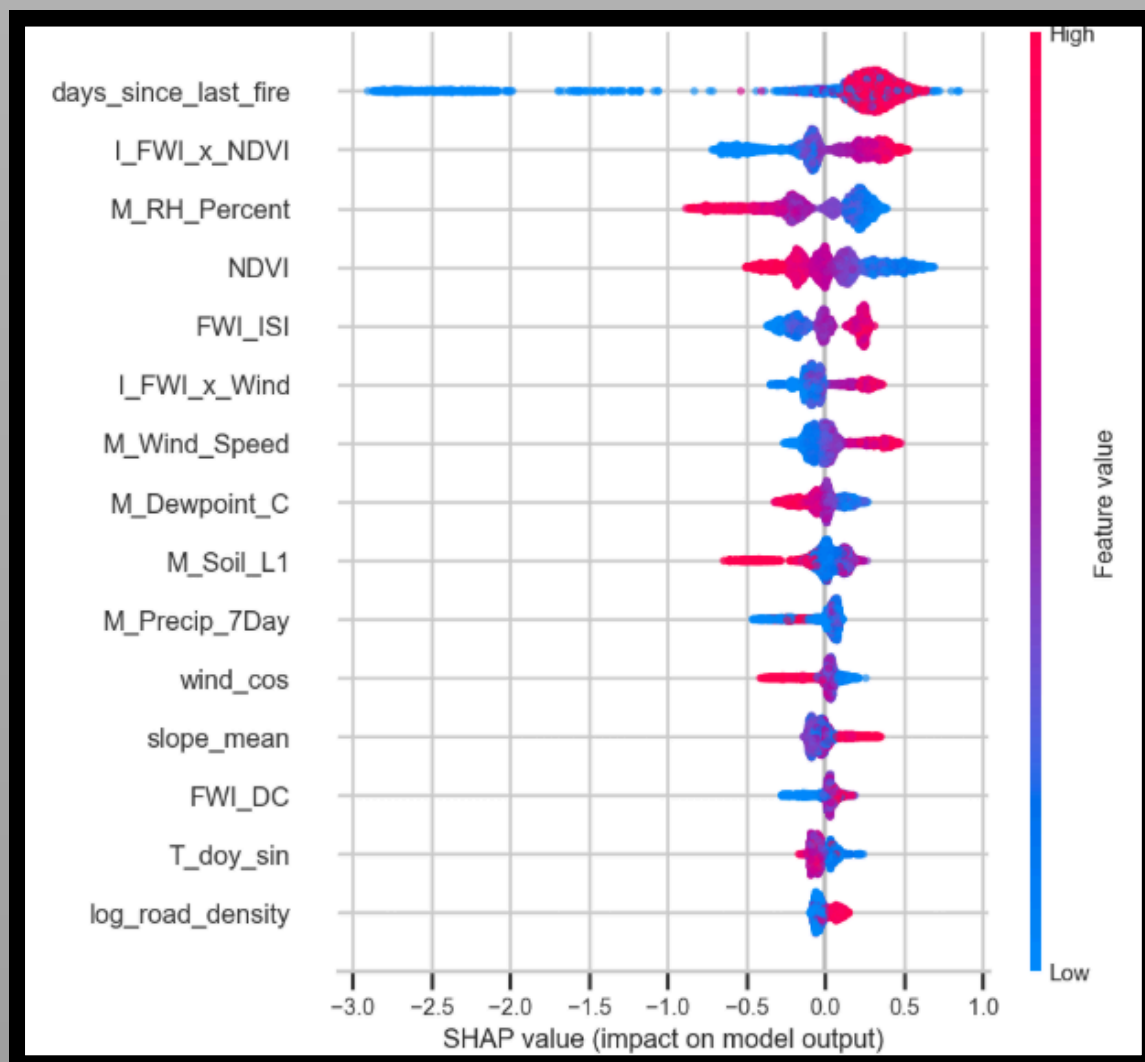


ALAZ with ERA5 Data

4.3. Explainable AI: Regional Feature Importance (SHAP Analysis)

Tree Gain and SHAP analysis confirmed the necessity of regionalization, revealing entirely different primary fire triggers per region:

- Mediterranean (MED):** The interaction term I_FWI×NDVI (22.3% importance) dominates. Fire risk peaks exclusively when extreme weather (FWI) aligns perfectly with dense, highly stressed vegetation (NDVI).
- Continental (CON):** FWI_Final (18.7%) and deep soil FWI_DC (9.7%) dictate the risk. Prolonged late-summer drought dynamics over fragmented forests are the primary triggers.
- Black Sea (BLK):** Driven heavily by FWI_FFM (15.4%) and VPD (8.9%). Being generally humid, fires here are triggered by rare, short-lived atmospheric vapor anomalies rapidly desiccating fine surface fuels.
- Universal Driver:** days_since_last_fire ranked consistently in the Top 5 most critical features (~4%) across all regions, tracking the localized accumulation of untreated forest fuel loading.



SHAP Analyze of MED Region

You can choose one of the three analyses.



FWI with ERA5 Data

CONCLUSION & REFERENCES

The foundational success of ALAZ lies in proving that a nation cannot be modeled uniformly for wildfire risk. Biogeographical regionalization captures precise, localized triggers—such as the Black Sea's sensitivity to rapid vapor pressure anomalies versus the Mediterranean's reliance on protracted vegetation distress. Secondly, the statistical confirmation that anthropogenic modeling (roads, settlements, fire history) significantly boosts predictive capacity underscores that fire is a human-environmental interaction, not purely a climatic occurrence.

By engineering a robust anti-leakage temporal split, training on physically coherent negative samples, and calibrating outputs via Platt Scaling, the system transforms theoretical ML scores into operationally dependable risk percentages (Brier Score < 0.088). Ultimately, this research transcends the theoretical scope by delivering a complete, high-performance web Decision Support System. Visualizing both traditional EFFIS FWI standards and the ALAZ machine learning engine simultaneously over 235K H3 grids, it bridges the gap between academic research and deployable forest management software.

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